WORKED SOLUTION

# STEP I 2003, Question 6

Pure • Integration by Partial Fractions

**STEP I 2003, Question 6**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q6

## The key idea

Both quadratics factorise, so the natural tool is partial fractions. The "different cases" come from the moment the two linear factors coincide: then partial fractions break down and we integrate a perfect square instead. Look hard at the two quadratics before starting part (ii): a simple shift of variable turns one into the other.

## Solution

### Part (i)

The quadratic factorises, since $2 \times a = 2a$ and $2 + a = a + 2$:

$$x^2 + (a+2)x + 2a = (x+2)(x+a).$$

Because $a > 0$, both factors are positive for $0 \le x \le 1$, so the integrand is continuous on the whole interval and the integral exists for every $a > 0$. The two factors are equal precisely when $a = 2$, and that is the dividing line.

**Case $a \neq 2$.** The factors are distinct, and

$$\frac{1}{(x+2)(x+a)} = \frac{1}{a-2}\left(\frac{1}{x+2} - \frac{1}{x+a}\right),$$

as we can check by recombining: the numerator becomes $(x+a) - (x+2) = a - 2$. Since both $x+2$ and $x+a$ are positive on $[0,1]$, we need no modulus signs in the logarithms:

$$\begin{aligned}\int_0^1 \frac{\mathrm{d}x}{(x+2)(x+a)} &= \frac{1}{a-2}\Big[\ln(x+2) - \ln(x+a)\Big]_0^1 \\ &= \frac{1}{a-2}\Big(\ln 3 - \ln(1+a) - \ln 2 + \ln a\Big) \\ &= \frac{1}{a-2}\ln\left(\frac{3a}{2(a+1)}\right).\end{aligned}$$

**Ask yourself** When $0 < a < 2$ the factor $\frac{1}{a-2}$ is negative. Is the answer still positive, as it must be for a positive integrand?

Yes. We have $\dfrac{3a}{2(a+1)} < 1 \iff 3a < 2a+2 \iff a < 2$, so for $0 < a < 2$ the logarithm is negative too, and the quotient is positive. Likewise for $a > 2$ both are positive. If you prefer every piece positive, write the answer for $0 < a < 2$ as $\dfrac{1}{2-a}\ln\left(\dfrac{2(a+1)}{3a}\right)$; it is the same number.

**Case $a = 2$.** Now the integrand is $(x+2)^{-2}$, and

$$\int_0^1 \frac{\mathrm{d}x}{(x+2)^2} = \left[-\frac{1}{x+2}\right]_0^1 = -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}.$$

#### Part (ii)

Here the quadratic is $u^2 + au + a - 1 = (u+1)(u+a-1)$, but before repeating the work, substitute $u = x+1$:

$$(x+1)^2 + a(x+1) + a - 1 = x^2 + (a+2)x + 2a.$$

This is exactly the quadratic of part (i). Also $\mathrm{d}u = \mathrm{d}x$, and $u = 1, 2$ correspond to $x = 0, 1$. So

$$\int_1^2 \frac{\mathrm{d}u}{u^2 + au + a - 1} = \int_0^1 \frac{\mathrm{d}x}{x^2 + (a+2)x + 2a},$$

and the answers are the same as in part (i):

$$\int_1^2 \frac{\mathrm{d}u}{u^2 + au + a - 1} = \begin{cases} \dfrac{1}{a-2}\ln\left(\dfrac{3a}{2(a+1)}\right), & a \neq 2, \\ \dfrac{1}{6}, & a = 2. \end{cases}$$

Directly: on $[1,2]$ we have $u + 1 > 0$ and $u + a - 1 \geq a > 0$, so the integrand is again continuous, and the factors coincide exactly when $a - 1 = 1$, that is $a = 2$. Partial fractions give $\frac{1}{a-2}\left[\ln\frac{u+1}{u+a-1}\right]_1^2 = \frac{1}{a-2}\left(\ln\frac{3}{a+1} - \ln\frac{2}{a}\right)$, which is the same expression.

## Checks

- **Continuity at $a = 2$.** As $a \to 2$, the logarithm tends to $\ln 1 = 0$, so the formula is of the form $0/0$. Its limit is the derivative of $\ln 3a - \ln(2a+2)$ at $a = 2$, namely $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, which agrees with the special case. The two cases join up smoothly, as they should.
- **A special value.** With $a = 1$ the integrand of (i) is $\frac{1}{(x+1)(x+2)}$ and the formula gives $-\ln\frac{3}{4} = \ln\frac{4}{3} \approx 0.2877$.
- **Numerically.** Simpson's rule on both integrals, for $a = 1$, $a = 2$, $a$ close to 0 and 2, and 200 random values of $a$ up to 20, agrees with the formula to better than $10^{-10}$.

## Common slips

- Forgetting the case $a = 2$, where the partial-fraction coefficient $\frac{1}{a-2}$ is undefined.
- Getting the sign of the partial fractions the wrong way round. Recombine to check.
- Treating $0 < a < 2$ and $a > 2$ as genuinely different answers. One formula covers both, provided you keep track of signs.

## Beyond STEP

The behaviour at $a = 2$ is a small example of a removable singularity: a formula that looks undefined at a point, but whose limit there is exactly the value of the integral. In analysis you learn when an integral depending on a parameter is continuous (or differentiable) in that parameter, which explains why the two cases had to agree. Bartle and Sherbert, *Introduction to Real Analysis*, treats limits, continuity and the Riemann integral carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.